Integral domain notes (From the Michael Penn Playlist)

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1 The field of fractions of an integral domain.

Motivation: We can think $\mathbb{Q}$ as a set of all symbols $\frac{a}{b}$ such that $a, b \in \mathbb{Z} \quad (b \neq 0)$ where

$$\frac{a}{b} = \frac{c}{d} \iff ad = bc$$

i.e

$$\frac{1}{2} = \frac{2}{4} \quad \text{because} \quad 1 \cdot 4 = 2 \cdot 2$$

Let D be an integral domain, and consider the set S

$$S = \{(a, b) \mid a, b \in D, b \neq 0\}$$

And define a relation $\sim$ on S where

$$\sim \subseteq S \times S, (a, b) \sim (c, d) \iff ad = bc$$

And we also define an equivalence class $[a, b]$ where

$$[a, b] = \{(c, d) \in S \mid (a, b) \sim (c, d)\}$$

Next we want to consider the set of all equivalence classes

$$F_D = \{[a, b] \mid a, b \in D, b \neq 0\}$$

We want to show that F_D is a field (the field of fractions of D) and it's the unique "smallest" field such that $D \hookrightarrow F_D$ First we want to show that $\sim$ is a equivalence relation

- Reflexive:
$$ab = ab \implies (a, b) \sim (a, b)$$
- Symmetry:
$$\begin{aligned} (a, b) \sim (c, d) &\implies ad = bc \\ &\implies bc = ad \\ &\implies (c, d) = (a, b) \end{aligned}$$
- Transitive:
Suppose $(a, b) \sim (c, d)$ and $(c, d) \sim (e, f)$. Then
$$\begin{aligned} ad = bc \quad cf = de \\ adf = bcf \quad bcf = bde \\ \implies adf = bde \\ \implies adf = bde \\ \implies (af)d = (be)d \\ \implies (a, f) \sim (b, e) \quad \text{(we can cancel because } d \neq 0) \end{aligned}$$

Next define addition in F_D.

$$[a, b] + [c, d] = [ad + bc, bd]$$

We then want to prove that addition is well defined. Suppose $[a, b] = [\hat{a}, \hat{b}]$ and $[c, d] = [\hat{c}, \hat{d}]$ meaning

$$(\hat{a}b = a\hat{b}) \quad (\hat{c}d = c\hat{d})$$

then we get

$$[a,b]+[c,d]=[ad+bc,bd]$$
$$[\hat{a},\hat{b}]+[\hat{c},\hat{d}]=[\hat{a}\hat{d}+\hat{b}\hat{c},\hat{b}\hat{d}]$$

In order to show that the RHS of both equations are equal to each other, we can cross multiply to see if they are equivalent. We can do this by our equivalence relation. Thus we want to show that

$$(ad+bc)(\hat{b}\hat{d})=(\hat{a}\hat{d}+\hat{b}\hat{c})(bd)$$

$$\begin{aligned}(ad+bc)(\hat{b}\hat{d}) &= ad\hat{b}\hat{d}+bc\hat{b}\hat{d}\\ &= \hat{a}bd\hat{d}+b\hat{b}\hat{c}d\\ &= (bd)(\hat{a}\hat{d}+\hat{b}\hat{c})\\ &= (\hat{a}\hat{d}+\hat{b}\hat{c})(bd)\end{aligned}$$

Thus our equivalence relation is well defined.
Next we want to show that F_D is a field such that $D \hookrightarrow F_D$:
Consider the map $\iota : D \to F_D$ where

$$\iota(a)=[a,1]$$

We want to show that it's a ring homomorphism. Note that it's a homomorphism under addition because

$$\iota(a+b)=[a+b,1]=[a,1]+[b,1]=\iota(a)+\iota(b)$$

It also preserves multiplication because

$$\iota(ab)=[ab,1]=[a,1][b,1]=\iota(a)\iota(b)$$

Thus it's a ring homomorphism. Next suppose that $\iota(a)=0$. Thus $a \in \ker\iota$. Meaning $[a,1]=[0,1]$ and we see that $\ker\iota=\{0\}$. Since the kernel of a homomorphism is trivial, then we see ι is injective. Thus $D \hookrightarrow F_D$.
Finally we want to show that F_D is the smallest such field.
Let k be a field such that $D \hookrightarrow_{\varphi} K$ where φ is an injective homomorphism. Thus we want to find $\psi : F_D \to k$ such that

$$D \hookrightarrow F_D \hookrightarrow k$$

Where $\psi \circ \iota = \varphi$. Set $\psi([a.b])=\varphi(a)\varphi(b)^{-1}$. We see that $a,b \in D$ and $\varphi(a),\varphi(b) \in k$. We need to show that ψ is an injective field homomorphism.

$$\begin{aligned}\psi([a,b]+[c,d]) &= \psi([ad+bc,bd])\\ &= \varphi(ad+bc)\varphi(bd)^{-1}\\ &= (\varphi(a)\varphi(d)+\varphi(b)(c))\varphi(b)^{-1}\varphi(d)^{-1}\\ &= \varphi(a)\varphi(b)^{-1}+\varphi(c)\varphi(d)^{-1}\\ &= \psi[a,b]+\psi[c,d]\end{aligned}$$

$$\begin{aligned}\psi([a,b]\cdot[c,d]) &= \psi([ac,bd])\\ &= \varphi(ac)\varphi(bd)^{-1}\\ &= \varphi(a)\varphi(c)\varphi(b)^{-1}\varphi(d)^{-1}\\ &= \psi[a,b]\psi[c,d]\end{aligned}$$

Thus it's a ring homomorphism (an injective ring homomorphism between fields is a field homomorphism) and we need to show ψ is injective.

Suppose that $[a,b] \in \ker \psi$. Thus $\psi[a,b] = 0$ and $\varphi(a)\varphi(b)^{-1} = 0$ Since φ is injective, then $\varphi(b) \neq 0$ and thus $\varphi(a) = 0$. Meaning $a = 0$ and thus $\psi[0,b] = 0$ but we know that in $[0,b] = [0,1]$ in D. Thus the kernel is trivial, and ψ is injective. Finally we want to show $\psi \cdot \iota = \varphi$. Evaluate $\psi \circ \iota(a)$ for some $a \in D$.

$$\begin{aligned} \psi \circ \iota(a) &= \psi[a,1] \\ &= \varphi(a)\varphi(1)^{-1} \\ &= \varphi(a) \end{aligned}$$

Thus if there exists a homomorphism from $D \to k$ then it can always be factored through the field of fractions, making F_D the unique "smallest" field that D is included inside of.

2 Irreducibles and Primes in Integral Domains

Let R be a commutative ring with multiplicative identity 1 and D be an integral domain.

- Suppose $a, b \in R$, then $a \mid b$ if $\exists c \in R$ such that $b = ac$
- We say that a and b are associates if $\exists$ a unit such that $a = ub$
- A non-unit $p \in D$ is irreducible if when $p = ab$ then a or b is a unit.
- p is a prime if when $p \mid ab$ then $p \mid a$ or $p \mid b$

Consider the following sub-ring of $\mathbb{Q}[x, y]$ generated by

$$R = \langle x^2, y^2, xy \rangle$$

We see that each generator is irreducible in R by xy is not prime.

$$xy \mid x^2y^2 \quad \text{but } xy \nmid x^2 \text{ and } xy \nmid y^2$$

Giving an example of something that is irreducible but also not prime. Consider the ring

$$\mathbb{Z}[i\sqrt{3}] = \{a + bi\sqrt{3} : a, b \in \mathbb{Z}\}$$

and the norm function $N : \mathbb{Z}[i\sqrt{3}] \to \mathbb{N}$ where $N(a + bi\sqrt{3}) = a^2 + 3b^2$. In the video he proves the following

- $N(x) = 0 \iff x = 0$
- $N(xy) = N(x)N(y)$
- u is a unit $\iff N(u) = 1$
- If $N(x)$ is a prime x is irreducible

The first 3 proofs are pretty easy so here's the proof for the last one.
Suppose that x is reducible and $N(x)$ is prime. Then $x = yz$. Applying the norm function $N(x) = N(y)N(z)$. Since $N(x)$ is prime then $N(y) = 1$ or $N(z) = 1$. Choosing $N(y) = 1$ we see y must be a unit. Meaning that x has to be prime. We also want to show that $1 + i\sqrt{3}$ is irreducible but not prime.
Suppose that we can factor $1 + i\sqrt{3} = xy$. Then

$$N(x)N(y) = N(1 + i\sqrt{3}) = 4$$

If x or y is a unit then we're done. If they are not units, then 4 must factor into 2×2 and we get

$$N(x) = N(y) = 2$$

Meaning if $x = a + b\sqrt{3}i$ then $a^2 = 2$ but $\sqrt{2} \notin \mathbb{Z}$. Thus x or y have to be a unit and thus $1 + i\sqrt{3}$ is irreducible. Next Since $4 = (1 + i\sqrt{3})(1 - i\sqrt{3})$ then

$$1 + i\sqrt{3} \mid 2 \times 2$$

but

$$1 + i\sqrt{3} \nmid 2$$

Meaning it's not prime. Thus we have another example of a ring with an element that is irreducible and not prime.

3 Introduction to Unique Factorization Domains (UFDs)

We say that an integral domain D us a UFD if

- Every nonzero, non-unit can be written as the product of irreducibles
- If $a = p_1 \cdot p_2 \cdot ... \cdot p_r = q_1 \cdot q_2 \cdot ... \cdot q_s$ where p_i are irreducible, then $r = s$ and $\exists \sigma \in S_r$ with $p_i = q_{\sigma(i)} u_i$, where u_i is a unit. (meaning every element in the q list is an associate of an element of the p list)

Recall that if $p \in D$ is irreducible if when $p = ab$ then a or b is a unit.
Some examples include

- $\mathbb{Z}$ is a UFD by the fundamental theorem of arithmetic.
- $k[x]$. The ring of polynomials with entries in a field k
- $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$

A non example is $\mathbb{Z}[i\sqrt{3}]$ because

$$4 = 2 \cdot 2 = (1 + i\sqrt{3})(1 - i\sqrt{3})$$

Thus $2 = (1 + i\sqrt{3})u$ where u is a unit. The video continues to prove there is no such unit in $\mathbb{C}$.

One more non example is $\mathbb{Z}[\sqrt{5}]$ since

$$4 = 2 \cdot 2 = (1 + \sqrt{5})(-1 + \sqrt{5})$$

similarly there exists no such unit such that $2 = (1 + \sqrt{5})u$

4 Introduction to Principal Ideal Domains (PIDs)

An integral domain D is called a PID if every ideal is principal. Recall that

- For all $a \in D$ the principal ideal generated by a is

$$\langle a \rangle = \{ra : r \in D\}$$

- a non-unit $p \in D$ is said to be irreducible if when $p = ab$, a or b is a unit
- a non-unit $p \in D$ is said to be a prime if when $p \mid ab$ then $p \mid a$ or $p \mid b$

Let D be a PID with $a, b \in D$. Then

1. $a \mid b \iff \langle b \rangle \subseteq \langle a \rangle$
2. a and b are associates $\iff \langle a \rangle = \langle b \rangle$
3. a is a unit $\iff \langle a \rangle = D$

Suppose $a \mid b$, then $b = ar$ for some $r \in D$. Let $x \in \langle b \rangle$ then $x = by$ for some $y \in D$. Then $x = (ar)y = a(ry)$. Thus $x \in \langle a \rangle$
Next suppose $\langle b \rangle \subseteq \langle a \rangle$. Then $b \in \langle a \rangle$ and thus $b = ar$ for some $r \in D$. Thus $a \mid b$

Suppose a, b are associates. Then there is a unit in D such that $a = bu$ AND $b = au^{-1}$. Thus $b \mid a$ AND $a \mid b$. Thus by the first proof $\langle a \rangle \subseteq \langle b \rangle$ and $\langle b \rangle \subseteq \langle a \rangle$. Thus

$$\langle a \rangle = \langle b \rangle$$

The converse of the proof is the exact same but in reverse

Suppose a is a unit, then a^{-1} exists. Take $x \in D$, then

$$x = (xa^{-1})a \in \langle a \rangle$$

Thus $D \subseteq \langle a \rangle$. But ideals are already sub-rings so $\langle a \rangle \subseteq D$. Thus $\langle a \rangle = D$.
Next suppose $D = \langle a \rangle$. Then $1 \in \langle a \rangle$. Thus there exists $b \in D$ such that $1 = ab$. Thus $b = a^{-1}$. Since the inverse of a exists, then a is a unit.

Theorem: Let D be a PID and $0 \notin \langle p \rangle \subseteq D$, then $\langle p \rangle$ is a maximal ideal $\iff$ p is irreducible.

Suppose $\langle p \rangle$ is a maximal ideal and $p = ab$. Then $a \mid p$ and thus $\langle p \rangle \subseteq \langle a \rangle \subseteq D$. But since $\langle p \rangle$ is a maximal ideal, then

$$\langle p \rangle = \langle a \rangle \text{ or } \langle a \rangle = D$$

Meaning:
p and a are associates and thus b is a unit, or a is a unit. In both cases p is irreducible.

Suppose that p is irreducible and consider $a \in D$ such that

$$\langle p \rangle \subseteq \langle a \rangle \subseteq D$$

Thus $a \mid p$ and hance $p = ab$ for some $b \in D$. Thus a or b is a unit. If a is a unit then $\langle a \rangle = D$. If b is a unit, then $\langle p \rangle = \langle a \rangle$. Thus by definition of a maximal ideal $\langle p \rangle$ is a maximal ideal.

As a corollary, we saw that in an arbitrary ring, prime implies irreducibility. In a PID, if $p \in D$ is irreducible, then p is prime.

Suppose p is irreducible and $p \mid ab$ then $ab = pr$ for some $r \in D$. Thus $ab \in \langle p \rangle$. Since $\langle p \rangle$ is a maximal idea, it must also be a prime ideal (R/M is a field and thus an integral domain). Thus since $ab \in \langle p \rangle$ then $a \in \langle p \rangle$ or $b \in \langle p \rangle$. Thus $p \mid a$ or $p \mid b$

4.1 Noetherian Rings

There isn't a video on Noetherian Rings but I'll include a small section on them since they are important for the next proof.

A Ring is Noetherian if it satisfies either of the following

1. The ring satisfies the acceding chain condition (ACC) if for every set of ideal $\{I_k\}_{j=1}$ such that

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq ...$$

   There exists $N \in \mathbb{N}$ such that for all $n \geq N$ then

$$I_n = I_N$$

2. Every ideal is finitely generated (Thus making every PID a Noetherian Ring)

I want to show that these statements are equivalent.
Proof: "2 $\implies$ 1"
Let R be a ring where every ideal is finitely generated. Take an acceding chain of rings such that

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq ...$$

Consider the ideal

$$I = \bigcup_{n=0}^{\infty} I_n$$

Then we want to show that I is an ideal. Take any $x, y \in I$, thus there exist $i, j \in \mathbb{N}$ such that $x \in I_i$ and $y \in I_j$. Let $m = \max(i, j)$. Thus $x, y \in I_m$ and $x + y \in I_m$. Thus $x + y \in I$ so it's closed under addition. Since I is an ideal in R, it must be finitely generated. Thus

$$I = \langle x_1, x_2, ..., x_n \rangle$$

Let n_i be the index such that $x_i \in I_{n_i}$. Let $N = \max\{x_i\}$. Thus for all $x \in I$ then $x \in I_N$. Meaning $I_N \subseteq I$ and $I \subseteq I_N$. Thus

$$I_N = I$$

Proof: "1 $\implies$ 2"
Suppose we have a ring that satisfies the ACC. Then assume for contradiction that there is an ideal I that is not finitely generated. For some element $x_1 \in I$ we see that $\langle x_1 \rangle \neq I$. Thus there exists $x_2 \in I$ that is not in $\langle x_1 \rangle$. We can keep going since I has infinitely many generators we can construct an infinite set of nested ideals strictly smaller than the last

$$\langle x_1 \rangle \subsetneq \langle x_1, x_2 \rangle \subsetneq \langle x_1, x_2, x_3 \rangle \subsetneq ...$$

But each sequence of nested ideals must terminate to finitely many generators. A contradiction. Thus I is finitely generated.

5 Every PID is a UFD

The definition of the ACC was brought up and then Michael Penn proceeded to prove that a PID satisfied the ACC similarly to how I proved that a Noetherian Ring satisfies the ACC and then a PID is a Noetherian Ring.

We want to show that Every PID has a factorization property and uniqueness of that factorization (thus being a UFD).
Let D be a PID. Suppose $a \in D$ that is non-zero and non-unit. Write $a = a_1 b_1$ where a_1 is not a unit. Then write $a_1 = a_2 b_2$ where a_2 is not a unit. Continuing this we write $a_k = a_{k+1} b_{k+1}$ where a_{k+1} is not a unit. Meaning

$$a_1 \mid a, a_2 \mid a_1, ..., a_{k+1} \mid a_k$$

Recall that $a \mid b \iff \langle b \rangle \subseteq \langle a \rangle$. Thus we get that

$$\langle a \rangle \subseteq \langle a_1 \rangle \subseteq \langle a_2 \rangle \subseteq \langle a_3 \rangle \subseteq ...$$

Since D satisfies the ACC, then there exists N such that for all $n \geq N$ then $\langle a_N \rangle = \langle a_n \rangle$. Meaning a_N and a_n are associates for all $n \geq N$. Thus the only way to factor a_N is by $a_N = pu$ where p is irreducible and u being a unit. Thus for all $a \in D$ we can factor out an irreducible $a = p_1 x_1$ where x is something else. Repeating this process we can write $x_1 = p_2 x_2$ where p_2 is irreducible, $x_2 = p_3 x_3$ where p_3 is irreducible, $x_k = p_{k+1} x_{k+1}$ where p_{k+1} is irreducible. We can create an ideal chain like the following

$$\langle x_1 \rangle \subseteq \langle x_2 \rangle \subseteq \langle x_3 \rangle \subseteq ...$$

Thus we have a terminating sequence starting with $\langle x_N \rangle = \langle x_{N+1} \rangle ...$ We then see that x_N is irreducible. We then see that $x_{N-1} = p_N x_N$. Thus x_{N-1} is a product of irreducibles. Set $x_N = p_{N+1}$. Thus we can write

$$a = p_1 x_1 = p_1 p_2 x_2 = p_1 p_2 p_3 ... x_N = p_1 p_2 p_3 ... p_{N+1}$$

Thus we took a non-zero unit and proved we can factor it into a product of irreducibles. Now we need to prove uniqueness of each factorization. Suppose there were two factorizations of some non-zero element $a \in D$

$$a = p_1 p_2 p_3 ... p_r = q_1 q_2 q_3 ... q_s$$

Thus we want to show that $r = s$ and $p_i = u_j q_j$ for some permutation if the indexing.

Assume that $r \neq s$ and WLOG assume $r < s$. We know that $p_i \mid a$ and thus $p_1 \mid q_1 q_2 ... q_s$. Since p_1 is prime then there exists q_j such that $p_1 \mid q_j$. We will re-index the product such that $p_1 \mid q_1$. Thus $q_1 = u_1 p_1$. Substituting in for q_1 we see

$$p_1(p_2 p_3 ... p_r) = u_1 p_1 (q_2 q_3 ... q_s)$$

Canceling p_1

$$p_2 p_3 ... p_r = u_1 q_2 q_3 ... q_s$$

Completing the same process for p_2 we see $q_2 = u_2 p_2$ and substituting again

$$p_2(p_3 ... p_r) = u_1 u_2 p_2 (q_3 ... q_s)$$

Repeat until we cancel all such p_i. We took such that $r < s$ so the LHS will cancel before the RHS. We will then get

$$1 = u_1 u_2 ... u_r q_{r+1} ... q_s$$

Thus we see that for all q_j for $j \in \{r+1, ..., s\}$ are units. But units are not irreducibles. A contradiction. Thus $r = s$. And by the same argument we also showed that $p_i = u_i q_i$ for some re-indexing of i. Hence factorizations of all non-zero elements of a PID are unique up to associates. By definition we proved that

Every PID is a UID

6 Introduction to Euclidean Domains

An integral domain D is a Euclidean domain if there exists $N : D \to \mathbb{N}$ such that

1. If $0 \neq a, b \in D$ then $N(a) \leq N(ab)$
2. If $a, b \in D$ where $b \neq 0$ then there exists $q, r \in D$ such that $a = bq + r$ where $r = 0$ or $N(r) < N(b)$

Some examples are $\mathbb{Z}$ where $N(m) = |m|$ and $k[x]$ where $N(f(x)) = \deg f(x)$. Note that $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ is a euclidean domain.
Proof: Define a norm $N(\alpha) = \alpha\overline{\alpha} = |\alpha|^2$ where

$$\alpha = a + bi \implies \overline{\alpha} = a - bi \implies |\alpha|^2 = a^2 + b^2$$

Checking each property

1. $\alpha, \beta \in \mathbb{Z}[i]$ then $N(\alpha\beta) = \alpha\beta\overline{\alpha\beta} = N(\alpha)N(\beta) \geq N(\alpha)$
2. Suppose $\alpha, \beta \in \mathbb{Z}[i]$ where $\beta \neq 0$. Where $\alpha = a + bi$ and $\beta = c + di$. Then we see that

$$\beta^{-1} = \frac{1}{\beta} = \frac{\overline{\beta}}{|\beta|^2} = \frac{c - di}{c^2 + d^2}$$

Thus we get $\alpha\beta^{-1} =$

$$\frac{(a + bi)(c - di)}{c^2 + d^2} = \frac{1}{c^2 + d^2}((ac + bd) + i(bc - ad))$$
$$= \frac{ac + bd}{c^2 + d^2} + \frac{bc - ad}{c^2 + d^2}i$$

We then want to take each rational and write them as integers plus some remainder. So we take the above equation to be

$$= (q_1 + r_1) + (q_2 + r_2)i$$

Thus we can always do this for $r_1, r_2 \in [-\frac{1}{2}, \frac{1}{2}]$ and for integers q_1, q_2. Thus

$$\alpha\beta^{-1} = (q_1 + r_1) + (q_2 + r_2)i$$

Rewriting we get

$$= q_1 + q_2 i + r_1 + r_2 i$$

We see that $(q_1 + q_2 i) \in \mathbb{Z}[i]$ and we'll set it equal to γ. Multiplying by β on both sides.

$$\alpha = \beta\gamma + \beta(r_1 + r_2 i)$$

Since α and $\beta\gamma$ are elements of $\mathbb{Z}[i]$ then $\beta(r_1 + r_2 i) \in \mathbb{Z}[i]$ and we will rewrite it as ρ. Thus

$$\alpha = \beta\gamma + \rho$$

. Taking the norm of ρ we get

$$N(\rho) = \beta\overline{\beta}(r_1 + r_2 i)(r_1 - r_2 i) = N(\beta)(r_1^2 + r_2^2)$$

Since we forced $-\frac{1}{2} \leq r_1, r_2 \leq \frac{1}{2}$ then we get that $r_1^2 + r_2^2 \leq \frac{1}{2}$ Thus

$$N(\rho) = N(\beta)(r_1^2 + r_2^2) \leq \frac{1}{2}N(\beta) < N(\beta)$$

6.1 Every Euclidean domain is a PID

Suppose $I \subset D$ is an ideal. (We want to show $I = \langle a \rangle$. We want to take $b \in I$ such that $N(b)$ is minimal among all elements from I. Thus $\langle b \rangle \subseteq I$. Take $a \in I$ then find q and r such that $a = bq + r$ where $r = 0$ or $N(r) < N(b)$. We know $r = a - bq$ and $a, bq \in I \implies a - bq \in I \implies r \in I$. But we chose b to be object in I such that $N(b)$ is the smallest. Meaning $N(r) < N(b)$ must be false. Thus $r = 0$ and thus $a = bq$ and thus $a \in \langle b \rangle$. Meaning $I \subseteq \langle b \rangle$. Thus every ideal $I \subset D$ is a principal ideal. Hence

Every Euclidean domain is a PID

7 If D is a UFD then $D[x]$ is a UFD.

Let D be an integral domain and F the field of fractions.

- $p(x) = a_n x^n + ... + a_0 \in D[x]$ is an example polynomial in x with coefficients in our field D.
- $Content(p(x)) = \gcd(a_0, ...a_n)$
- $p(x)$ is primitive is the content $= 1$

There are a few lemmas we will need.

- **Gauss' Lemma:** If $f(x), g(x) \in D$ are primitive, then $f(x)g(x)$ is primitive.
- **Multiplicative Content:** $Cont(f(x)g(x)) = Cont(f(x))Cont(g(x))$
- **Factorization Lemma:** Suppose $p(x) \in D[x]$ where $p(x) = f(x)g(x) \in F[x]$, then there exist $\hat{f}(x), \hat{g}(x) \in D[x]$ such that $p(x) = \hat{f}(x)\hat{g}(x)$
- **Corollary:** $p(x)$ is irreducible in $D[x] \iff p(x)$ is irreducible in $F[x]$

Proof of Gauss' Lemma:
Assume that $f(x), g(x) \in D$ are prime but $f(x)g(x)$ is not. Then there exists a prime number that divides every coefficient of $f(x)g(x)$. Take $p \in D$ to be such a prime. Since $f(x), g(x)$ are prime, then p does not divide all of their coefficients. Take r is the smallest coefficient such that $p \nmid a_r$ and let s be the smallest such that $p \mid b_s$. Let c_{r+s} be the coefficient of the x^{r+s} term in $f(x)g(x)$. We get that the coefficient is

$$c_{r+s} = a_{r+s}b_0 + a_{r+s-1}b_1 + ... + a_0 b_{r+s}$$

Regrouping the terms we get

$$c_{r+s} = (a_{r+s}b_0 + ... + a_{r+1}b_{s-1}) + a_r b_s + (a_{r-1}b_{r+1} + ... + a_0 b_{r+s})$$

We know that $p \mid c_{r+s}$ and p divides the first half because we choose s to be the smallest and everything has indexing smaller than s and likewise with the right half and p.

$$\underbrace{c_{r+s}}_{p \text{ divides this}} = \underbrace{(a_{r+s}b_0 + ... + a_{r+1}b_{s-1})}_{p \text{ divides this}} + a_r b_s + \underbrace{(a_{r-1}b_{r+1} + ... + a_0 b_{r+s})}_{\text{and } p \text{ divides this}}$$

But we choose r, s such that $p \nmid a_r$ and $p \nmid b_s$. Thus $p \nmid a_r b_s$ a contradiction. Thus there is no prime p that divides every coefficient.

Proof of multiplicative content:
Set $f(x) = a\hat{f}(x)$ where $a = Cont(f(x))$. Thus $\hat{f}(x)$ is primitive.
Similarly set $g(x) = b\hat{g}(x)$ where $b = Cont(g(x))$. Thus $\hat{g}(x)$ is primitive. Thus

$$f(x)g(x) = ab\hat{f}(x)\hat{g}(x)$$

By the previous lemma we know that $\hat{f}(x)\hat{g}(x)$ is primitive. We then see that $Cont(f(x)g(x)) = ab$ and thus

$$Cont(f(x)g(x)) = Cont(f(x))Cont(g(x))$$

Factorization Lemma
Suppose $p(x) \in D[x] \hookrightarrow F[x]$. Where $p(x) = f(x)g(x)$ where $f(x)g(x) \in F[x]$. Since the coefficients of $f(x)$ and $g(x)$ have denominators we want to cancel, then take $a, b \in D$ such that $af(x) \in D[x]$ and $bg(x) \in D[x]$. Next we want to factor out $\hat{a}, \hat{b} \in D$ such that

$$af(x) = \hat{a}\hat{f}(x) \quad bg(x) = \hat{b}\hat{g}(x)$$

where $\hat{f}(x), \hat{g}(x)$ are primitive. We see that

$$abp(x) = abf(x)g(x) = \hat{a}\hat{b}\hat{f}(x)\hat{g}(x)$$

Since $\hat{f}(x)$ and $\hat{g}(x)$ are primitive, we see that $ab \mid \hat{a}\hat{b}$ and so we get c such that $\hat{a}\hat{b} = c(ab)$. Substituting in we get

$$abp(x) = c(ab)\hat{f}(x)\hat{g}(x)$$

Canceling ab

$$p(x) = c\hat{f}(x)\hat{g}(x)$$

Finally **Proof if D is a UFD then $D[x]$ is a UFD**
Take $p(x) \in D[x] \hookrightarrow F[x]$ since $F[x]$ is a Euclidean domain, then it's also a UFD. Thus we can factor

$$p(x) = f_1(x)...f_n(x)$$

where $f_i(x)$ is irreducible in $F[x]$. Take $a_i \in D$ such that $a_i f_i(x) \in D[x]$. Thus we can cancel out the denominators in our field of fractions. We can then take $b_i \in D$ such that $a_i f_i(x) = b_i g_i(x)$ where $g_i(x) \in D[x]$ are all primitive. But by the Corollary, $g_i(x)$ is also irreducible in $D[x]$. Next consider the product of all a_i times $p(x)$

$$a_1a_2...a_np(x) = a_1f_1(x)a_2f_2(x)...a_nf_n(x) = b_1b_2...b_ng_1(x)g_2(x)...g_n(x)$$

Note the RHS is a product of primitive polynomials times some constants. We will short hand the product of a_i to a and b_is to b. Then we see that where $g_i(x)$ is primitive, then for $p(x)$ we have

$$ap(x) = bg_1(x)g_2(x)...g_n(x)$$

Meaning that the GCD of the coefficients of the LHS is 1. Meaning $a \mid b$ and thus $b = ac$ for some $c \in D$. Rewriting we get

$$ap(x) = acg_1(x)g_2(x)...g_n(x)$$

Note that since D is a UFD, then there exists unique set of irreducibles c_i such that $c_1c_2...c_m = c$. After canceling a we get

$$p(x) = c_1c_2...c_mg_1(x)g_2(x)...g_n(x)$$

Thus our polynomials can be factored into a product of irreducibles in $D[x]$.

Next we want to show uniqueness. Suppose there were two factorizations of $p(x)$ such that

$$p(x) = a_1a_2...a_mf_1(x)f_2(x)...f_n(x) = b_1b_2...b_rg_1(x)g_2(x)...g_s(x)$$

Viewing inside of $F[x]$ which is a PID $\implies$ UFD. Thus $n = s$. Since units from $F[x]$ are in F, the we can rearrange $g_i(x) = \dfrac{c_i}{d_i} f_i(x)$ for $c_i, d_i \in D$. Then clearing the denominator we get

$$d_ig_i(x) = c_if_i(x) \in D[x]$$

Since we could choose f_i and g_i to be primitive, then c_i and d_i are associates. Meaning there is a unit u_i such that $d_i = u_ic_i$. Rewriting we get

$$(a_1...a_m)(c_1...c_m)(f_1(x)...f_n(x)) = (b_1...b_r)(u_1...u_m)(c_1...c_m)(f_1(x)...f_n(x))$$

Since this is all happening in an integral domain, we can cancel large portions of the equation such as all the f and c terms. Then we get

$$(a_1...a_m) = (b_1...b_r)(u_1...u_m) \in D$$

Since D is a UFD, then $r = m$ and a_i and b_j are associates up to reordering. Since we also know that f_i and g_j are also associates up to reordering, then any factorization of $p(x)$ into irreducibles is unique up to associates.

8 Summary of Integral Domains

So far we learned that

$$\text{Field} \implies \text{Euclidean domain} \implies \text{PID} \implies \text{UFD} \implies \text{Integral domain} \implies \text{Comm Ring w/ 1}$$

And we want to show that these implications are strict. That is, there's a counter example to UFD $\implies$ PID and so forth.

1. A commutative ring that is not an integral domain

   $$\mathbb{Z}_{12}$$

   Since $3, 4 \in \mathbb{Z}_{12}, 3 \times 4 = 0$

2. An integral domain that is not a UFD

   $$\mathbb{Z}\left[i\sqrt{5}\right]$$

   Since $6 = 2 \times 3$ and also $6 = \left(1 + i\sqrt{5}\right)\left(1 - i\sqrt{5}\right)$

3. A UFD that is not a PID.

   $$\mathbb{Z}[x], \mathbb{Q}[x, y]$$

   Since $\mathbb{Z}$ is a UFD, so is $\mathbb{Z}[x]$. However $\langle x, 2\rangle$ is not a principle ideal.
   I thought I should include this since it wasn't mentioned in the video: $D[x]$ is a PID if and only if D is a field. $\mathbb{Z}$ is not a field since $2 \in \mathbb{Z}$ is not a unit.

4. A PID that is not a Euclidean Domain (explanation included much later)

   $$\mathbb{Z}\left[\frac{1}{2}\left(1 + i\sqrt{19}\right)\right]$$

5. A euclidean domain that is not a field

   $$\mathbb{Z}, k[x]$$

   Where k is a field

6. Fields that are nice :)

   $$\mathbb{Q}, \mathbb{R}, F_D, \mathbb{Z}_p$$

Next, recall important definitions for a non-zero/non-unit p. We say p is irreducible if when $p = ab$, then a or b is a unit. We say p is prime if when $p \mid ab$ then $p \mid a$ or $p \mid b$. Recall the implications in the kinds of Rings we have

1. Commutative Ring / Integral Domain

   $$\text{prime} \implies \text{irreducible}$$

   An example where an irreducible is not prime is in $\mathbb{Z}\left[i\sqrt{5}\right]$. We see that $\left(1 - i\sqrt{5}\right)\left(1 + i\sqrt{5}\right) = 6$ Meaning $\left(1 + i\sqrt{5}\right) \mid 6 \implies \left(1 + i\sqrt{5}\right) \mid 3 \times 2$. But

   $$\left(1 + i\sqrt{5}\right) \nmid 3 \quad \left(1 + i\sqrt{5}\right) \nmid 2$$

   Making it irreducible but not prime

2. UFD / PID / Euclidean Domain

   $$\text{prime} \iff \text{irreducible}$$

3. Field

   No irreducibles or primes (vacuously true)

8.1 Dedekind-Hasse Norm

Let D be an integral domain. A function $N : D \to \mathbb{N}$ is called a Dedekind-Hasse norm if for all $0 \neq a, b \in D$ then $b \mid a$ or there exists s, t such that $0 < N(sa - tb) < N(b)$. Consider the set $\tilde{D}$ where

$$\tilde{D} = \{\text{the set of units}\} \cup \{0\} = R^{\times} \cup \{0\}$$

$u \in \tilde{D}$ is called a universal side divisor if for all $x \in D$ then there exists $z \in \tilde{D}$ such that $u \mid x - z$. Meaning

$$x = qu + z, z \in \tilde{D}$$

First we want to show that every Euclidean domain has a universal side divisor
Let $u \in D \backslash \tilde{D}$ such that $N(u)$ is minimal among all such elements. Let $x \in D$. Do the division algorithm with x and u. Then

$$x = qu + r$$

with $r = 0$ or $N(r) < N(u)$. If $r = 0$ then $u \mid x$ meaning $u \mid x - 1$ and thus $0 \in \tilde{D}$. If $r \neq 0$ then by the minimality of u we know $r \in \tilde{D}$. In either case

$$u \mid x - r, r \in \tilde{D}$$

Thus u is a universal side divisor.

Next we want to show that D is a PID if and only if it has a Dedekind-Hasse norm.
Proof: " $\Longleftarrow$ "
Suppose D has a Dedekind-Hasse norm N. Let I be a non-zero ideal and $b \in I$ such that $N(b)$ is minimal. Suppose $a \in I$ and consider the sub-ideal $\langle a, b \rangle \subseteq I$. Suppose $b \mid a$ then $a = br, r \in D$ and thus $a \in \langle b \rangle$. If $b \nmid a$ then we have $s, t \in D$ such that

$$0 < N(sa - tb) < N(b)$$

Note that $sa - tb \in \langle a, b \rangle \subseteq I$ but that contradicts b being minimal. Thus $a \in \langle b \rangle$ for all $a \in I$. Thus $I \subseteq \langle b \rangle$ and $\langle b \rangle \subseteq I$. Meaning

$$I = \langle b \rangle$$

Hence, every ideal is principal and thus D is a PID.

Proof: " $\Longrightarrow$ "
Suppose D is a PID ($\Longrightarrow$ UFD). We define $N : D \to \mathbb{N}$ such that $N(0) = 0$, $N(u) = 1$ if $u \in D$ is a unit, $N(a) = 2^n$ of $a = p_1 p_2 ... p_n$ for $p_i \in D$ are irreducible (and thus prime). Note that

$$N(a)N(b) = N(ab)$$

Suppose $0 \neq a, b \in D$ and choose $r \in D$ such that $\langle a, b \rangle = \langle r \rangle$. We can do this because every ideal is principal in a PID. If $b \mid a$ then we satisfied the Dedekind-Hasse rule. If $b \nmid a$ and thus $a \notin \langle b \rangle$ and thus $r \notin \langle b \rangle$ (since $a \in \langle r \rangle$). So we see $\langle b \rangle \subsetneq \langle r \rangle$ meaning $b = rx$ for a non-unit x Thus

$$N(b) = N(r)N(x) > N(r)$$

Since $r \in \langle a, b \rangle$ then there exist $s, t \in D$ such that $r = sa - tb$. Substituting and taking norms we see

$$0 < N(r) = N(sa - tb) < N(b)$$

Thus we proved that if D is a PID $\Longleftrightarrow$ it has a Dedekind-Hasse Norm. Now to provide an example of a PID that is not a euclidean domain

9 A PID that is not a Euclidean Domain

Consider the ring $R = \mathbb{Z}[w]$ where

$$w = \frac{1}{2}\left(1 + i\sqrt{19}\right)$$

Our strategy is to define a Dedekind Hasse Norm and then show that there is no universal side divisor. Meaning it's a PID but not a Euclidean domain.
Define $N : R \to \mathbb{N}$ by

$$N(a + bw) = (a + bw)(a - bw) = (a^2 + ab + 5b^2)$$

Suppose that $\alpha, \beta \in R$ where $\beta \nmid \alpha$ meaning $\dfrac{\alpha}{\beta} \notin R$. Consider the inclusion into the field

$$R \hookrightarrow \mathbb{Q}\left[i\sqrt{19}\right]$$

Thus we know that β has an inverse in here and we define it as

$$\frac{\alpha}{\beta} = \frac{a + bw}{c} \quad c \neq 1 \quad \gcd(a, b, c) = 1$$

Thus there exists $x, y, z \in \mathbb{Z}$ such that $ax + by + cz = 1$. Doing the division algorithm with $ay - 19bx$ and c we have

$$ay - 19bx = cq + r$$

Since we're working with integers we can choose the smaller of the positive and negative remainders (r or $c - r$) and get one such that $|r| < \dfrac{c}{2}$.

Throwing numbers around now, let's set $s = y + xi\sqrt{19}$ and let $t = q - zi\sqrt{19}$. The next totally obvious (not really) identity for our ring $\mathbb{Q}\left[i\sqrt{19}\right]$ is

$$0 < N\left(\frac{\alpha}{\beta}s - t\right) = \frac{\overbrace{(ay - 19bx - cq)}^{<(\frac{c}{2})^2}{}^2 + 19\overbrace{(ax + by + cz)}^{=1}{}^2}{c^2} \leq \frac{1}{4} + \frac{19}{c^2}$$

Pushing the norm into our ring R

$$0 < N\left(\alpha s - \beta t\right) \leq N(\beta)\left(\frac{1}{4} + \frac{19}{c^2}\right)$$

Note that $N(\alpha s - \beta t) < N(\beta)$ if $c \geq 5$. Meaning we constructed $\alpha s - \beta t$ such that the norm is satisfied for $c \geq 5$. Thus it would suffice to check for $c = 2, 3, 4$. And I will just provide s, t that will suffice since computation and number crunching is very boring for me.

For $c = 2$

$$s = 1, \quad t = \frac{(a - 1) + bi\sqrt{19}}{2}$$

For $c = 3$. If $a^2 + 19b^2 = 3q + r$ then $r \in \{1, 2\}$

$$s = a - bi, \quad t = q$$

For $c = 4$ then a and b are not both even. Thus suppose one is even and one is odd. Then we have $a^2 + 19b^2 = 4q + r$ for $r \in \{1, 3\}$

$$s = a - bi\sqrt{19}, \quad t = q$$

If a and b are both odd then $a^2 + 19b^2 = 8q + 4$

$$s = \frac{1}{2}\left(a + bi\sqrt{19}\right), \quad t = q$$

Thus for every case R has a Dedekind-Hasse Norm. Meaning R is a PID (fucking FINALLY)

Next we want to show that R has not universal side divisors. The only units in R are ± 1. Thus

$$\tilde{R} = \{-1, 0, 1\}$$

Suppose $u \in R \backslash \tilde{R}$ is a universal side divisor. Let $N(a+bw) = a^2+ab+5b$ be the norm on $\mathbb{Q}\left[i\sqrt{19}\right]$. Not the smallest values of N are $1, 4$ since $N(\pm 1) = 1$ and $N(\pm 2) = 4$. Let $x = 2$. Then there is $z \in \tilde{R}$ where $u \mid x - z$

1. Suppose $z = 0$ then $u \mid x$. Since $x = 2$ then $u = \pm 1, \pm 2$

2. If $z = 1$ then $u \mid 1$ and then $u = \pm 1$

3. If $z = -1$ then $u \mid 3$ and then $u = \pm 3$

But $u \neq \pm 1$ since u is not a unit. Thus the only possible universal side divisors are $\pm 2, \pm 3$. Check $2 \nmid w - z$ and $3 \nmid w - z$ for $z \in \{0, \pm 1\}$. The cases are easy to check and thus we see there are no universal side divisors. Meaning that R is not a euclidean domain.

10 Ideals of quotients of PIDs

Fourth Isomorphism Theorem:
Every ideal $\overline{J} \subseteq R/I$ is of the form J/I where $I \subseteq J \subseteq R$
Zorn's Lemma:
Every Poset with the property that every increasing chain has a maximal element, has a maximal element for the whole poset.

Ex: Suppose D is a PID and $I \subseteq D$ is an ideal, then every ideal of D/I is principal. (Note: D/I may not be a PID since it might not be an integral domain). Suppose $\overline{J} \subseteq D/I$ is an ideal. Then since D is a PID, then we can write

$$J = \langle b \rangle$$

We claim that $\overline{J} = \langle b + I \rangle$. Clearly $\langle b + I \rangle \subseteq \overline{J}$ since $b + I \in \langle b \rangle + I$. Suppose $x \in \overline{J}$. Rewriting we see

$$x \in \frac{J}{I} = \frac{\langle b \rangle}{I} = \{i + J \mid j \in \langle b \rangle\}$$

Thus there exists $r \in D$ such that $x = rb + I$. Meaning $x = (r + I)(b + I)$ and thus $x \in \langle b + I \rangle$.

Corollary: If $P \subseteq D$ is a prime ideal, then D/P is a PID. Proof: We know by the example, that D/P is a principal ring, and D/P is an integral domain since P is a prime ideal. These are sufficient conditions for D/P to be a PID
Ex: If D is a PID, then every prime ideal is Maximal. Proof: Suppose $P = \langle p \rangle$ is a prime ideal and we have $P \subseteq I \subseteq D$. We want to show that $I = P$ or $I = D$. Let $I = \langle a \rangle$. Since $\langle p \rangle \subseteq \langle a \rangle \subseteq D$ then $a \mid p$. But P being a prime ideal means that p is irreducible. Thus a can either be a unit or an associate of p. If a is a unit, then $\langle a \rangle = D$. If a is an associate of p then $\langle a \rangle = \langle p \rangle$. Thus $\langle p \rangle = P$ is a maximal ideal.

Ex: Let D be an integral domain such that every prime ideal is principal, then D is a PID. Proof: Assume for contradiction that there exists a set of non-principal ideals

$$A = \{I \subseteq D \mid \text{where } I \text{ is not principal}\}$$

This is a poset with ordering $I \subseteq I'$. Consider the ideals $I_1, I_2 ... \in A$ where

$$I_1 \subseteq I_2 \subseteq I_3 ...$$

Consider the ideal

$$I_\infty = \bigcup_{i=1}^{\infty} I_i$$

Assume for contradiction that $I_\infty \notin A$ Then $I_\infty = \langle a \rangle$ for some $a \in D$. Then $a \in I_n$ for some n. Thus we see

$$\langle a \rangle \subseteq I_n \subseteq I_\infty = \langle a \rangle \implies I_n = \langle a \rangle$$

But we assumed that all $I_i \in A$ are not principal ideals. A contradiction. Thus $I_\infty \in A$. Thus we showed that A has a maximal element and call it I. Meaning any ideal strictly containing I is principal. We also know that I is not a prime ideal. Thus we have $a, b \in D$ such that $a \notin I$ and $b \notin I$. Thus

$$I \subsetneq \langle I, a \rangle = \langle \alpha \rangle$$

We can conclude something similar with b. Where $I \subsetneq \langle I, b \rangle$. And we will define a new ideal

$$J = \{r \in D \mid ra \in I\}$$

And we see $\langle I, b \rangle \subseteq J$ and by the maximality of I, J is also a principal ideal we will call represent as $\langle \beta \rangle$. Note that

$$x \in I \subseteq \langle \alpha \rangle \implies x = r\alpha, r \in D$$

Also notice that

$$r\langle\alpha\rangle \subseteq \langle x\rangle \subseteq I$$

Recall that $a \in \langle\alpha\rangle$. Meaning $ra \in I$. Then $r \in J = \langle\beta\rangle$ and thus for some y we get $r = y\beta$. Substituting into y then $x = y(\alpha\beta) \in \langle\alpha\beta\rangle$ and thus $I \subseteq \langle\alpha\beta\rangle$. Note that $\beta \in J$, then $\beta a \in I$. But since I is an ideal, then $i\beta \in I$. Thus for all $s \in \langle I, a\rangle$ we have $\beta s \in I$. In particular take $s = \alpha$. Meaning $\alpha\beta \in I$ and thus $\langle\alpha\beta\rangle \subseteq I$. Then $\langle\alpha\beta\rangle = I$. However this contradicts that I is the largest non-principal ideal. Thus D is a PID.

And that's a good place to stop.